Research Questions:

1. **"How does the user interface** *of food delivery apps* **like Swiggy influence users' food choices?"**
2. "What **other factors influence** users' food choices on food delivery apps like Swiggy?"

Motivation

The topic was chosen taking into consideration the growing reliance on food delivery apps among individuals. As convenience becomes a primary driver, many users **may be nudged towards unhealthy options through dark patterns** (such as visual cues, placement, promotions, or algorithmic recommendations).

This research is motivated by the concern that UI design may deliberately steer users away from healthier choices, shaping long-term eating habits in subtle but impactful ways.

We chose the app "Swiggy" as it is one of the most widely used food delivery apps in India.

Literature Review

8 papers (4- Dark Patterns, 4- OFD apps)

1. UI Manipulation & Dark Patterns

- 95% of apps use at least one dark pattern (AppRay, 2024)
- Users rarely recognize them (only 25% could spot)
- Patterns like urgency, prechecked boxes, default selections
- Normalized in routine app usage

- **From Exploration to Revelation (2024)**
- **What Makes a Dark Pattern... Dark? (2021)**

2. OFD & Unhealthy Food Choices

- Convenience, price, and visual appeal > nutrition
- Frequent app use linked to lower healthy food intake
- Users often order fast food (Jordan Study, 2022)

- **Determinants of OFD Choices (2025)**
- **Healthy Food & OFD Apps (2022)**

3. Research Gaps & Need

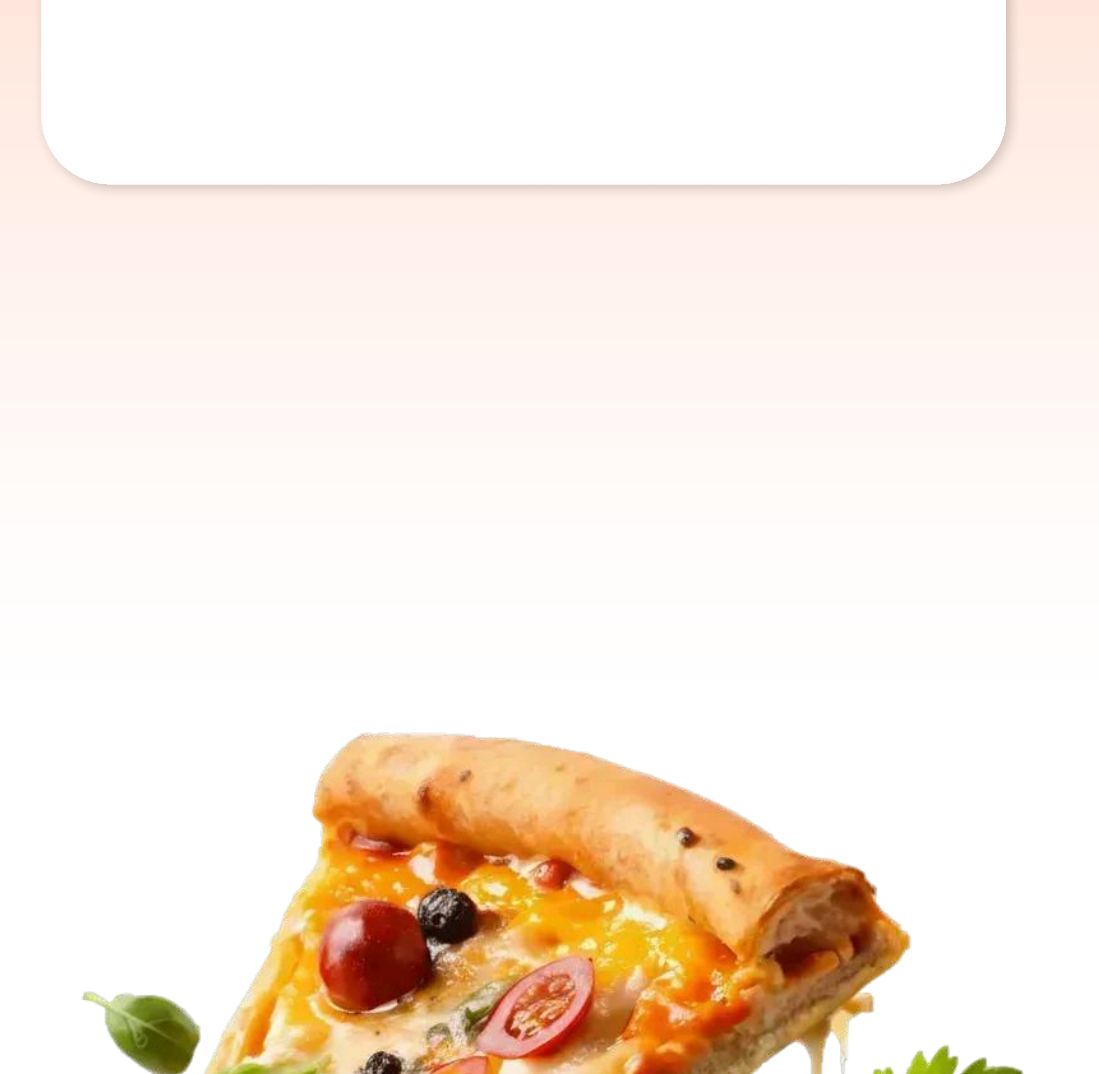
- Lack of focus on dark patterns + food decisions
- Limited research on students & working professionals
- Need to classify dark patterns specific to OFD platforms

- **Unintended Consumption (2023)**
- **UI Dark Patterns in Food Apps (2024)**

Data collection

12 Young adults (18 - 30) 7 Older adults (30 - 50)

Purposive sampling method
People who are well versed with using Swiggy.



Research Method

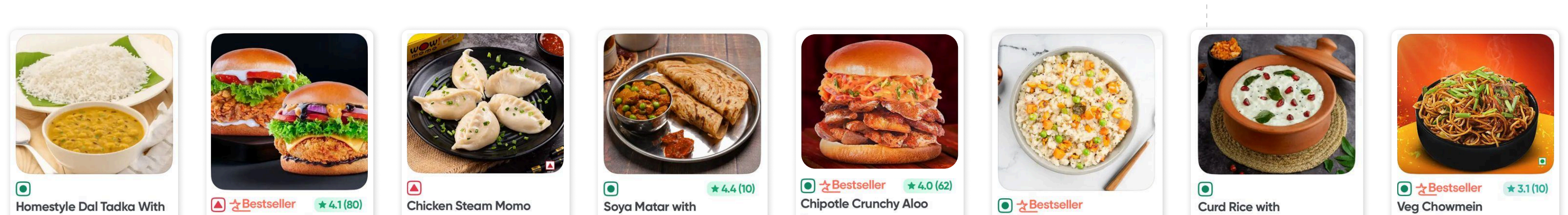
PHASE 1
App analysis - Swiggy :
- Analyzed food delivery app 'Swiggy' for 4-5 specific dark patterns that may influence user food choices.

PHASE 2
Task Based Scenario followed by semi-structured interview:
- Imagine you order food from the food delivery app "Swiggy" on a frequent basis (3-4 times a week). You want to order a meal today.
- **Task:** Order a meal on 'Swiggy'

PHASE 3
Collection of users' screen recordings while ordering a meal on 'Swiggy' followed by classification based on dark pattern influence identified via app analysis (whether their choices have been influenced/not).

PHASE 4
Eye tracking: Tracking users' gaze to test whether unhealthy food items' images are aesthetically manipulative using UI screen having 4 healthy and 4 unhealthy food options.

Participant selection
- 18-30 young adults - **10 participants** ***(5 males, 5 females)***



Homestyle Dal Tadka With Steamed Rice
~~₹215~~ ₹109 ADD

Bestseller ★4.1 (80)
Non-Veg Burger Duos
₹199 ADD

Chicken Steam Momo
~~₹179~~ ₹129 ADD

★4.4 (10)
Soya Matar with Paratha-2PC
~~₹182~~ ₹149 ADD

Bestseller ★4.0 (62)
Chipotle Crunchy Aloo Burger
₹79 ADD

Bestseller
Rava Upma
₹131 ADD

Curd Rice with Pomegranate
~~₹212~~ ₹149 ADD

Bestseller ★3.1 (10)
Veg Chowmein
~~₹179~~ ₹129 ADD

Data Analysis Method

PHASE 1 - *Identifying dark patterns*

- Performed a **Collaborative Walkthrough** where we explored the "Swiggy" app and identified the 4 major dark patterns using predefined dark patterns taxonomy (Gray et al.)

Price framing
Using financial incentives to encourage less healthy but more profitable choices.
99 store, combo deals, discount prices

Obstruction
Making the path to healthier food more effortful or less obvious than default browsing.
Algorithm, default restaurant/ food item line up

Deceptive promotion
Blurring the line between ads and personalized suggestions.
Recommended' items, food ad banners

Aesthetic Manipulation
Unhealthy food has flashy images; healthy options look dull.
Differences in healthy vs. unhealthy product images

PHASE 2 - *Identify the themes for both of our Research questions*

- For Phase 2 Data Analysis, we performed **Thematic Analysis** of the data that we got through our semi- structured interview transcripts.

PHASE 3 - *Screen Recording Analysis*

- For Phase 3 Data Analysis, we conducted a **Screen Recording Analysis with Interviews** of participants' ordering journeys, followed by **Deductive Classification** of dark patterns.

Insights such as items ordered, dark patterns identified, and daily eating habits, etc. were compiled into a structured spreadsheet for cross-participant comparison and pattern synthesis.

PHASE 4 - *Screen Recording Analysis*

- In Phase 4, we conducted **Gaze Count Analysis.** The goal was to identify whether unhealthy food items attracted more visual interest due to aesthetic manipulation.

Findings

RQ1 : How does the user interface of food delivery apps like Swiggy influence users' food choices?

Visuals strongly impact what users end up ordering
Users are drawn in by food images, especially aesthetic, tempting, or familiar ones.

Add-ons at checkout push users to expand orders
Low-cost or well-placed add-ons during checkout lead users to make impulse additions.

Notifications have limited but strategic influence
Notifications rarely drive decisions but can nudge users sometimes or when creatively executed.

Familiarity and discovery are shaped by app experience
Users rely on recognizable names, previous orders, and app-driven suggestions to discover or stick to certain restaurants.

Offers, combos and limited - time deals detour from original intent
Visible discounts, combos, and offer timers often override users' initial food choices.

Q2 : What factors influence users' food choices on food delivery apps like Swiggy?

Speed of delivery is consistently the top priority
Quick delivery is the most critical factor influencing food choice, often more than price or ratings.

Pricing, discount and cost related incentives drives most decisions
Price-sensitive users compare platforms; weak deals or extra fees reduce interest.

Cravings, impulses, and emotional connection to food drive spontaneous ordering
Familiar items and cravings, often mood-driven, can override planned choices.

Trust and familiarity with restaurants drive loyalty and limit exploration
Users stick to trusted places for hygiene and past positive experiences.

Food preferences follow a clear weekly rhythm
Weekdays favor healthy meals; weekends are for indulgent, preferred foods.

Meal timing, environment and weather shapes what gets ordered
Time, mood, company, and weather influence ordering and food choices.

Screen-recording + Interview Results (PHASE 3)

8/19 users' choices were influenced due to dark patterns.

2/7 of older adults got influenced by a dark pattern, whereas 6/12 younger adults got influenced by a dark pattern.

Obstruction
Affected **1/19** users

Price Rating
Affected **3/19** users

Aesthetic Manipulation
Affected **4/19** users

9 out of 19 **users ordered items that they don't eat on a usual/regular basis** despite the given scenario.

We infer that due to **mental association**, whenever **ordering online**, users often gravitate towards **unhealthy food** choices.

Unhealthy
7/19 participants

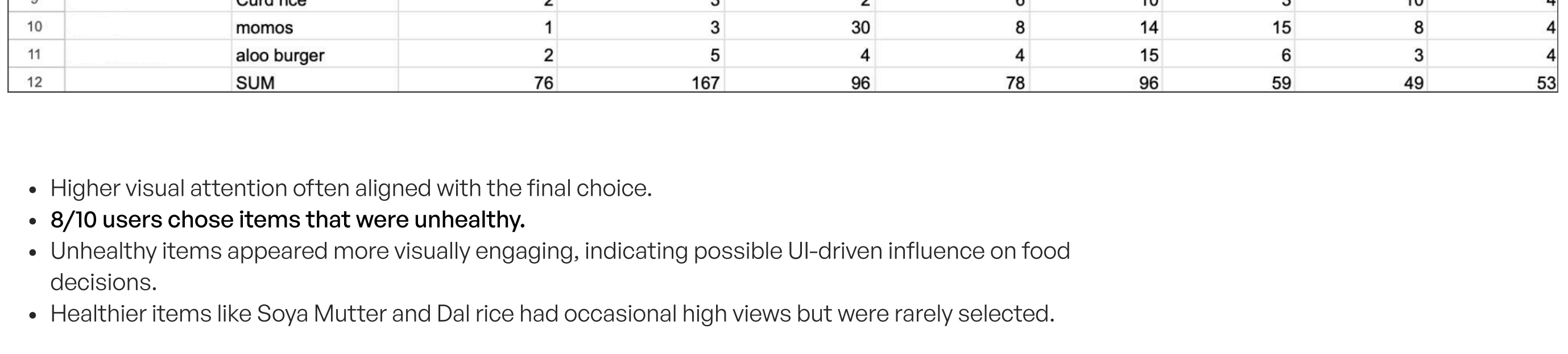
Moderately Unhealthy
6/19 participants

Moderately Healthy
4/19 participants

Healthy
2/19 participants

1 2 3 4 5 6 7 8 9 10

Eye-tracking Analysis Results (PHASE 4)



	A	B	C	D	E	F	G	H	I	J
1	Participant	chosen item	Dal rice	chicken burger duo	momos	soya mutter	aloo burger	rava upma	curd rice	chowmein
2		chicken burger duo	21	47	31	9	14	6	8	8
3		chicken burger duo	14	70	10	17	19	8	6	5
4		Dal rice	15	4	4	2	4	3	4	3
5		chicken burger duo	6	13	4	4	5	2	1	6
6		chicken burger duo	5	14	7	13	11	8	3	4
7		chicken burger duo	2	5	1	8	0	2	1	1
8		chowmein	8	3	3	7	4	6	5	14
9		Curd rice	2	3	2	6	10	3	10	4
10		momos	1	3	30	8	14	15	8	4
11		aloo burger	2	5	4	4	15	6	3	4
12		SUM	76	167	96	78	96	59	49	53

- Higher visual attention often aligned with the final choice.
- **8/10 users chose items that were unhealthy.**
- Unhealthy items appeared more visually engaging, indicating possible UI-driven influence on food decisions.
- Healthier items like Soya Mutter and Dal rice had occasional high views but were rarely selected.

Implications

Not all users are affected equally by dark patterns: Dark patterns affected only 8 of 19 users, with cravings or familiarity often being stronger influences.

The effect can be subtle: Dark patterns subtly nudge behavior, often boosting order value without being the main driver.

Mental association is a bigger driver: Users link online ordering with unhealthy food; dark patterns exploit this bias rather than create it.

Thank You